4. **Iteration:** Steps 2 and 3 are repeated until the centroids stabilize (i.e., they no longer change significantly) or the algorithm reaches a predefined number of iterations.

This iterative process ensures that the clustering improves with each iteration, leading to well-defined groupings.

Advantages of K-Means

- **Scalability:** K-Means performs efficiently on large datasets and can handle thousands or even millions of data points.
- **Simplicity:** The algorithm is easy to implement and interpret, making it accessible to both beginners and experienced practitioners.
- **Efficiency:** K-Means converges quickly for well-separated clusters, making it a preferred choice in many scenarios.

Limitations of K-Means

- **Assumptions of Cluster Shape:** K-Means assumes that clusters are spherical and of similar size, which may not hold for datasets with irregularly shaped or overlapping clusters.
- **Sensitivity to Outliers:** Outliers can disproportionately influence centroid calculations, leading to suboptimal clustering.
- **Choosing the Value of k:** Selecting the number of clusters (k) can be challenging and often requires domain knowledge or evaluation techniques like the Elbow Method or Silhouette Analysis.

Implementation in Python

Below is an example of implementing K-Means clustering in Python using the Scikit-Learn library: from sklearn.cluster import KMeans import pandas as pd

Example dataset